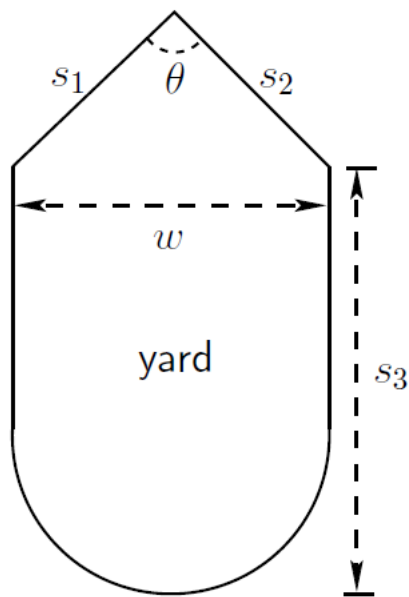


1 Overview

A local landscaper has a popular yard design he builds for customers that has an unusual shape (see diagram below, not exactly drawn to scale). Often he has to cover the yard with topsoil before planting. Write a C program that computes and reports, given the measured dimensions s_1 , s_2 , s_3 , angle θ (in degrees), and a desired depth of topsoil, the 1) area of the yard, 2) the yard's perimeter, and 3) the volume of topsoil needed to cover the yard.

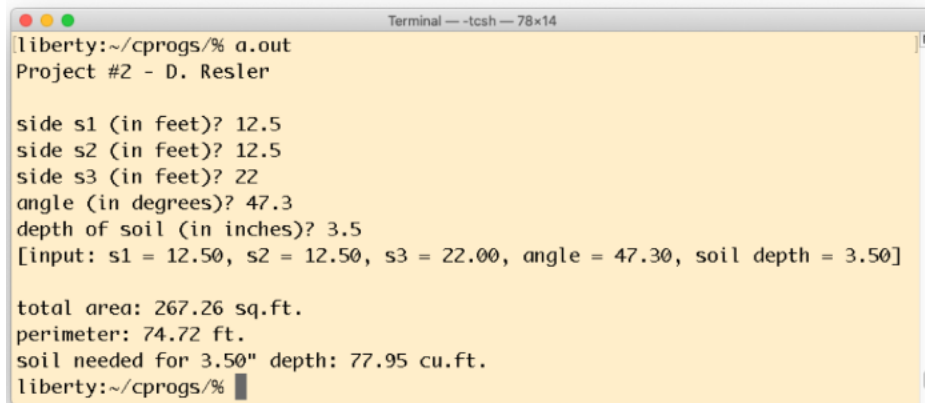


Note that $s_1 = s_2 > 0$, $s_3 \geq w/2$, and $45 \leq \theta \leq 120$. The curved portion of the yard is a complete semicircle, i.e. one half of a circle. You may assume that the sides of the center portion of the yard are parallel with each other and of equal length.

2 Implementation

In order for it to be more likely that everyone will get the same correct results, all variables in your program should be of type double and you must use, if needed, the built-in constant `M_PI` in the C library (just add `#include <math.h>` at the top of your program to get access to it and other math routines) for pi. In addition any math functions that you may use from the library must both accept and return double values. Your prompts plus output labels must match the sample run *exactly* (with the exception that you of course will use your name instead of Dr. Resler's and mine). Note that the output line that begins with '[input:]' is just an echo print of the input data. Also note that while the input data is suppose to be within a certain range, you may just assume that it is correct as we have not yet covered how you might check for this.

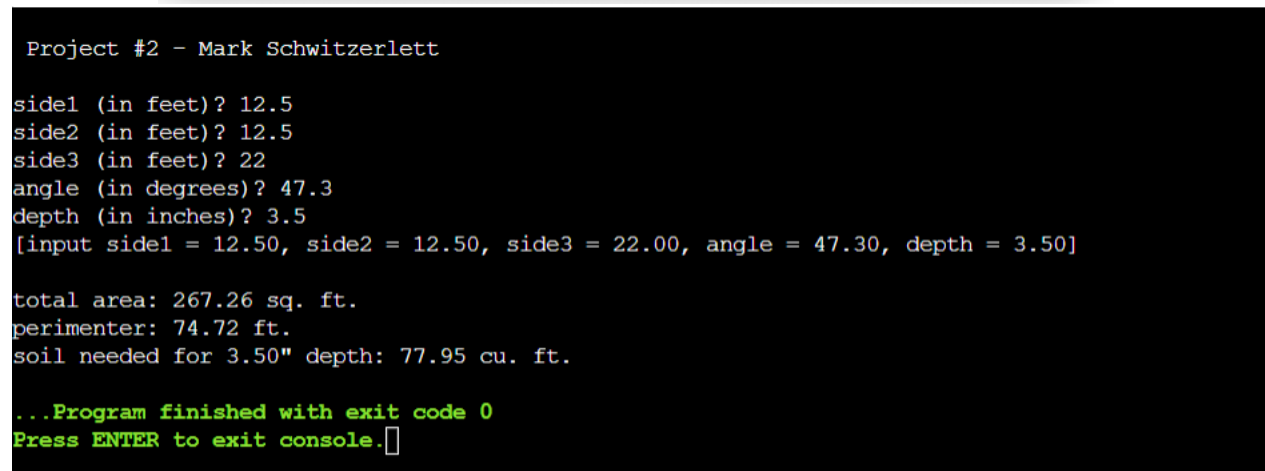
3 Sample Run



```
liberty:~/cprogs/% a.out
Project #2 - D. Resler

side s1 (in feet)? 12.5
side s2 (in feet)? 12.5
side s3 (in feet)? 22
angle (in degrees)? 47.3
depth of soil (in inches)? 3.5
[input: s1 = 12.50, s2 = 12.50, s3 = 22.00, angle = 47.30, soil depth = 3.50]

total area: 267.26 sq.ft.
perimeter: 74.72 ft.
soil needed for 3.50" depth: 77.95 cu.ft.
liberty:~/cprogs/%
```



```
Project #2 - Mark Schwitzerlett

side1 (in feet)? 12.5
side2 (in feet)? 12.5
side3 (in feet)? 22
angle (in degrees)? 47.3
depth (in inches)? 3.5
[input side1 = 12.50, side2 = 12.50, side3 = 22.00, angle = 47.30, depth = 3.50]

total area: 267.26 sq. ft.
perimeter: 74.72 ft.
soil needed for 3.50" depth: 77.95 cu. ft.

...Program finished with exit code 0
Press ENTER to exit console.
```

4 Deliverables

You should turn in a stand-alone, complete application program (your source code) containing a main function. Projects this term will be submitted via Gradescope. Make sure to document your code in the manner previously discussed in class. Note you need not turn in anything other than your source code text ie (the .c file).

DUE DATE: Monday, September 22, 11:59 pm